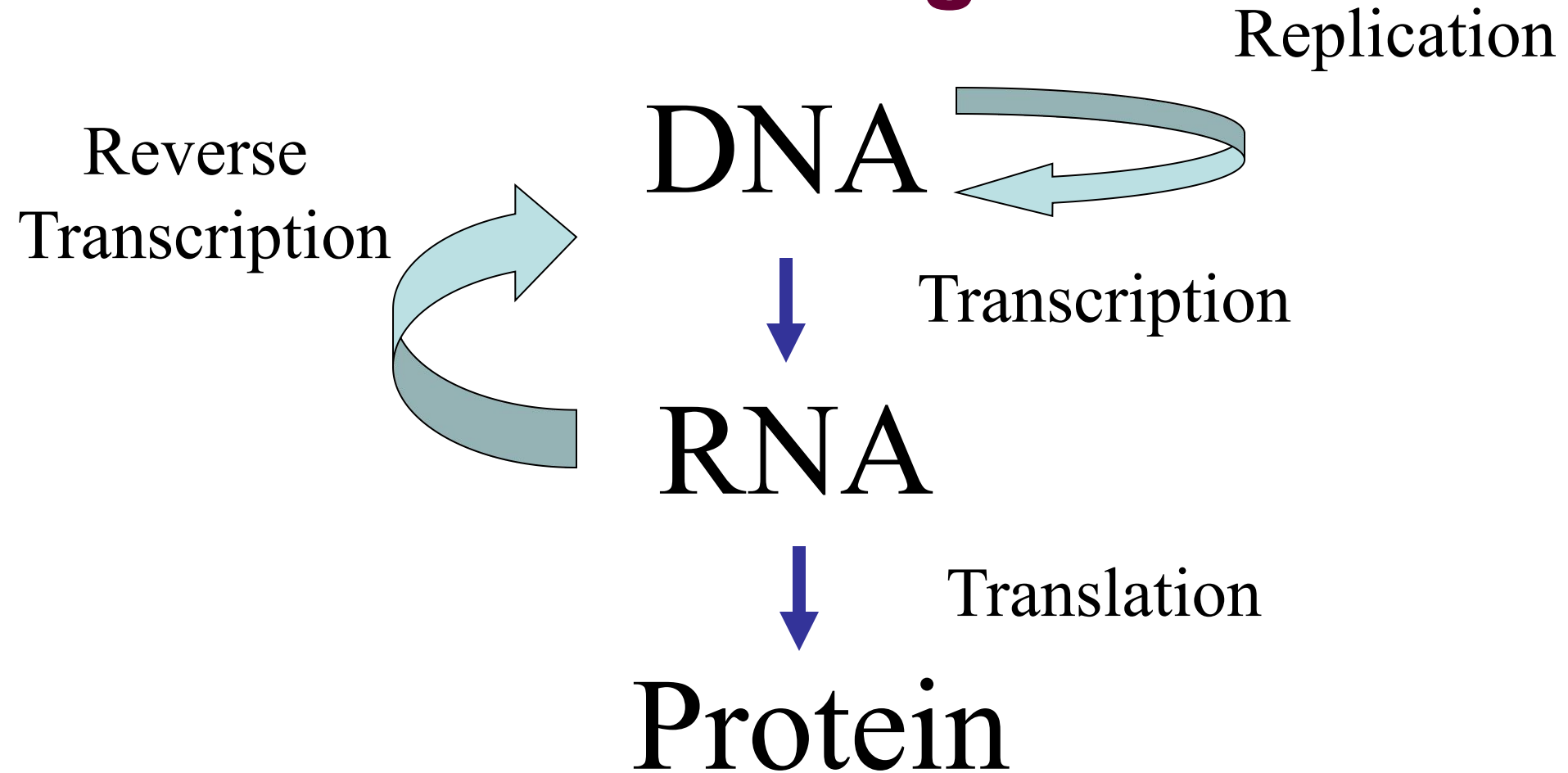


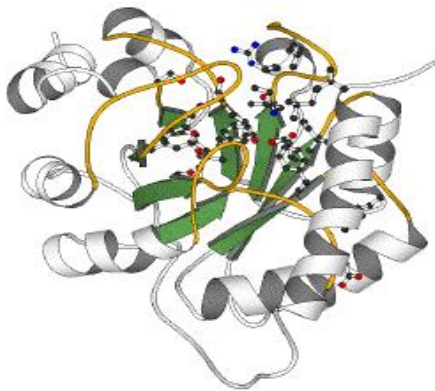
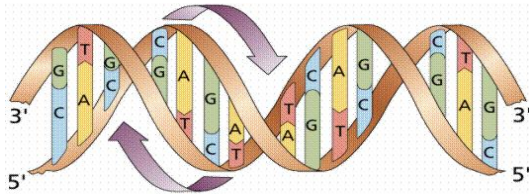
Chapter 2

The molecular nature of genes

Central Dogma



A gene codes for a protein



DNA

transcription

mRNA

translation

Protein

GGACTCGGTTGATAACTACTT
CCTGAGCCAACTATTGATGAA

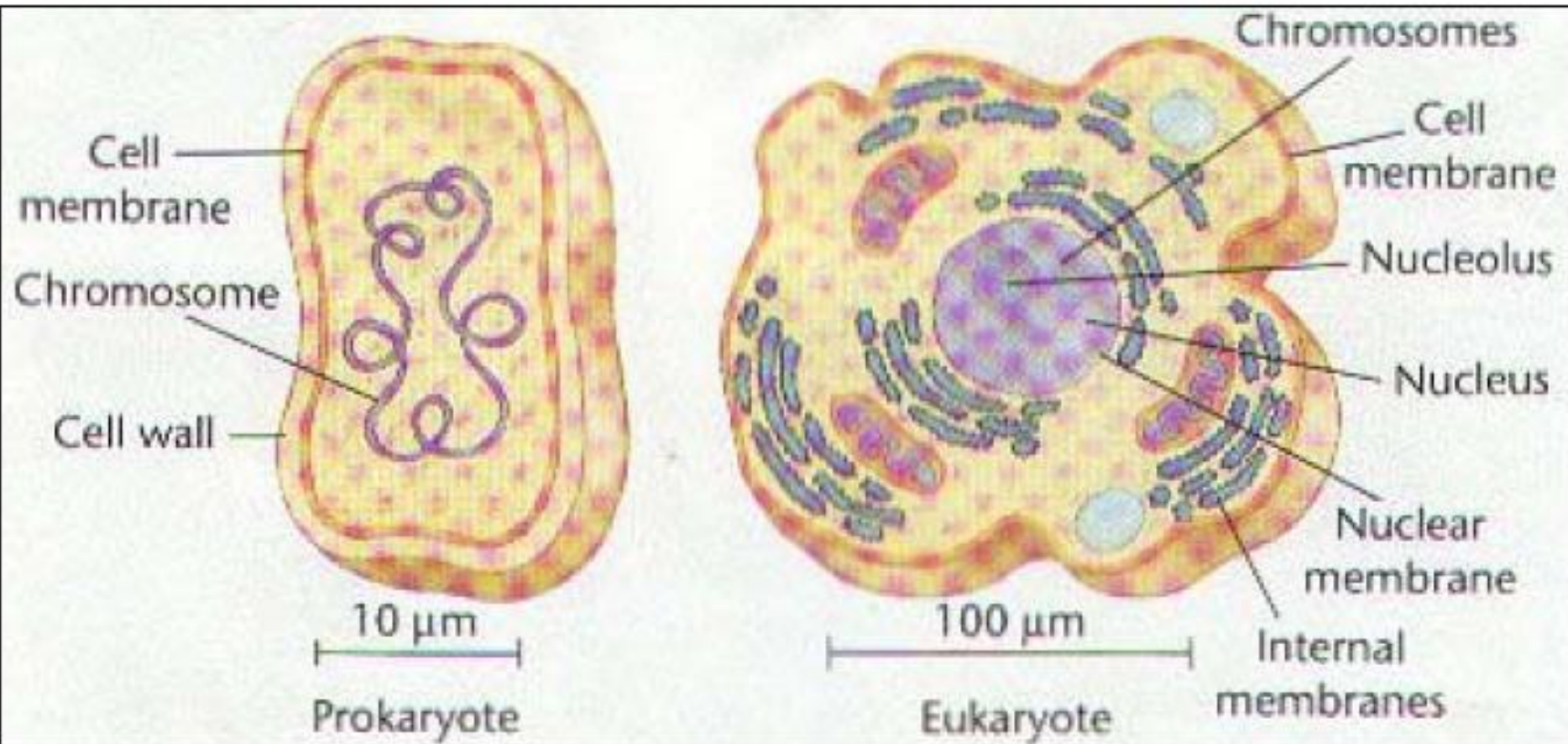


CCUGAGCCAAACUAUUGAUGAA

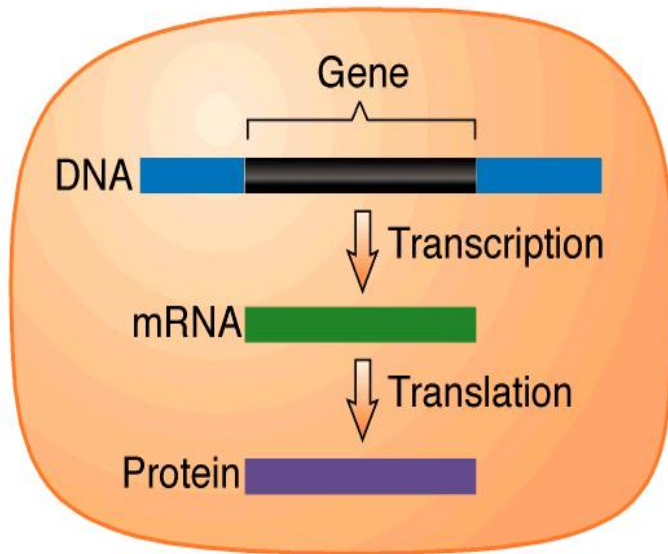


PEPTIDE

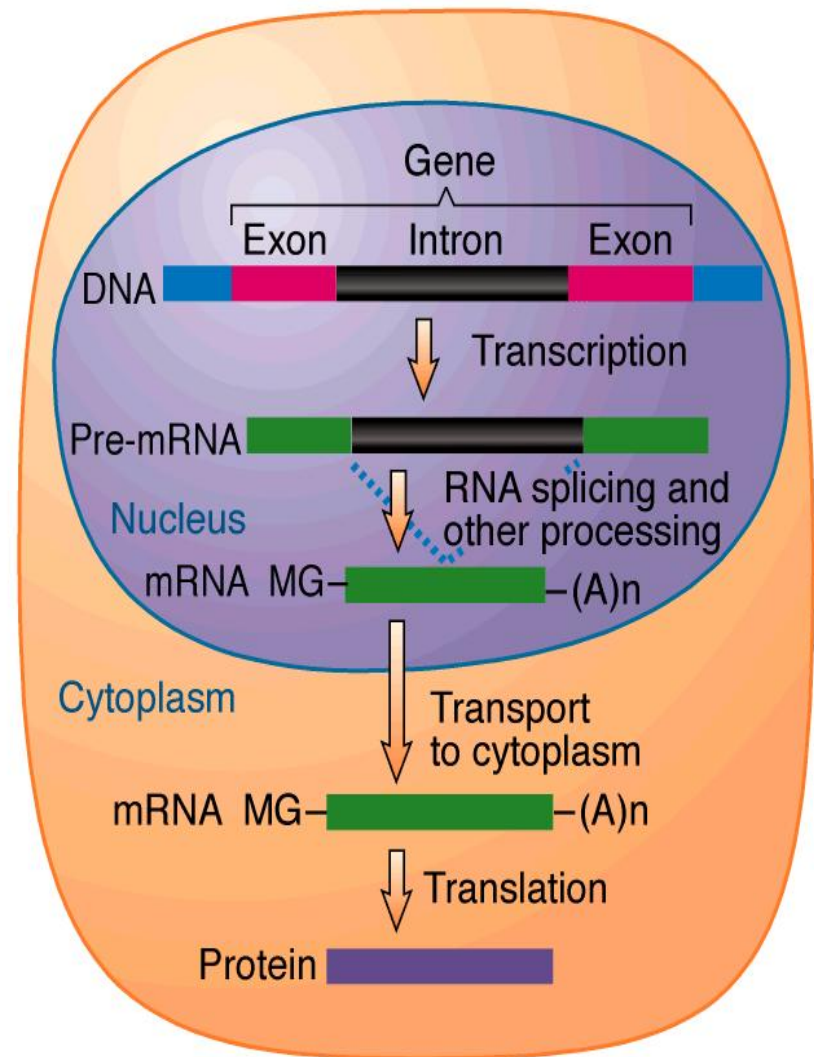
Prokaryotic vs. eukaryotic cell



The difference of gene expression between prokaryotes and eukaryotes



(a) Prokaryotes.



(b) Eukaryotes.

Prokaryote

Eukaryote

General:

Site of Transcription	Protoplasm	Nucleus
Site of DNA Replication	Protoplasm	Nucleus
Site of RNA Splicing	Rare	Nucleus
Site of Translation	Protoplasm	Cytoplasm
Site of Recombination	Protoplasm	Nucleus

DNA Replication:

Chromosomes	Circular, One Origin	Linear, Multiple Origins
Telomeres	No	Yes

Chemical nature of genes

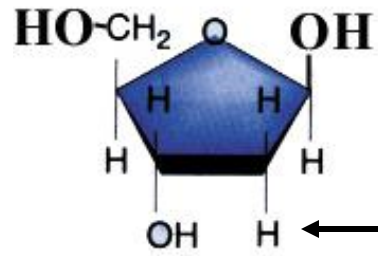
脱氧核糖核酸 DNA (Deoxyribonucleic Acid) & RNA (Ribonucleic Acid)

- **Polynucleotides** (DNA and RNA) contains:
(deoxy)ribose, phosphoric acid; bases
- **Bases includes:**

Two purines: adenine and guanine (腺嘌呤和鸟嘌呤)

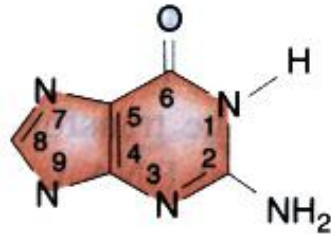
Three pyrimidines: cytosine, uracil, thymine (胞嘧啶、尿嘧啶和胸腺嘧啶)

- **DNA is the genetic material of organisms.**
- **In some viruses (HIV) RNA is used.**

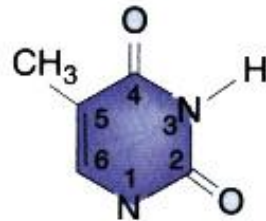


Sugar

OH group in ribonucleotide



Base



Phosphate

DNA

DNA

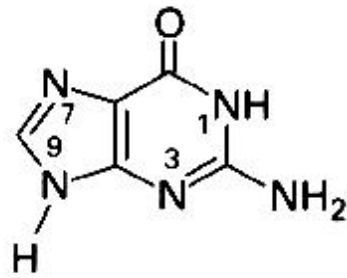
DNA

DNA

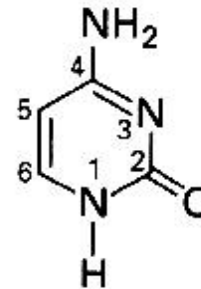
RNA



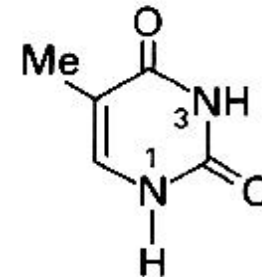
adenine



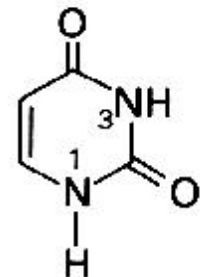
guanine



cytosine



thymine

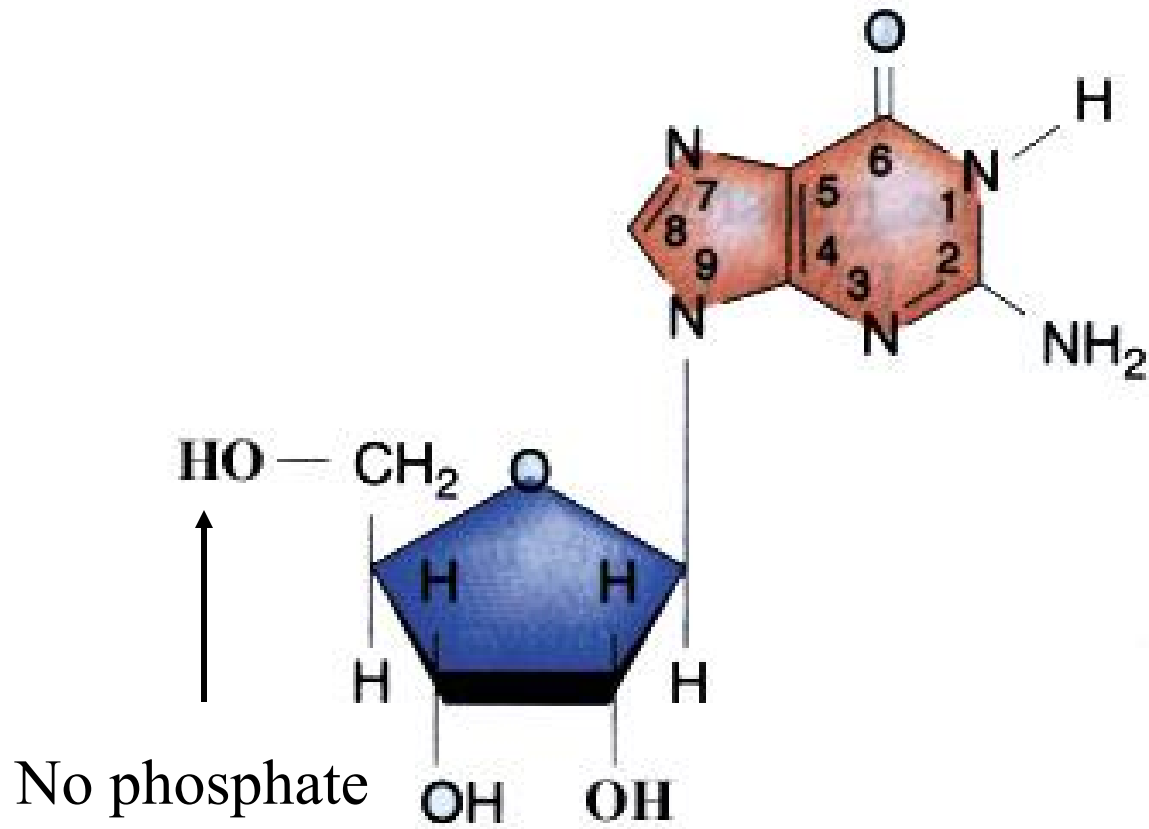


uracil

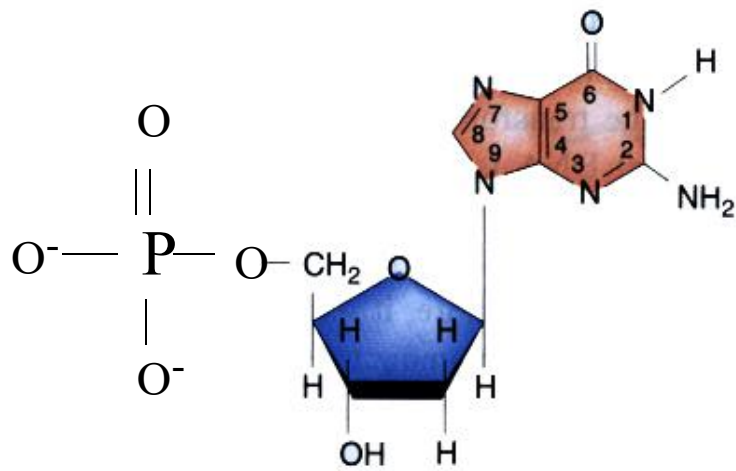
Purines

Pyrimidines

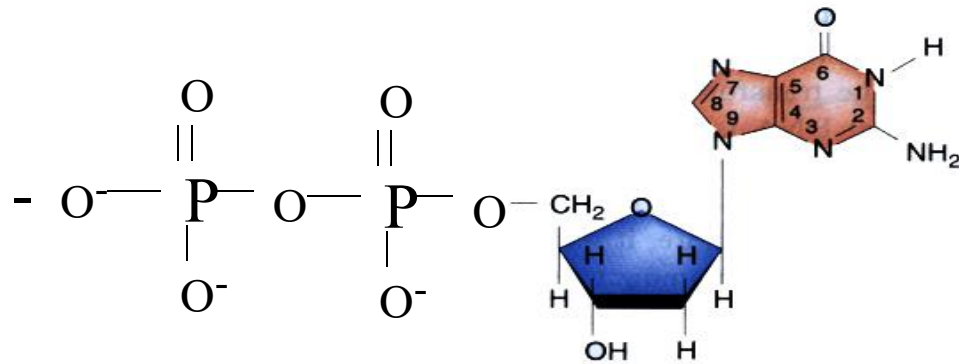
In RNA Uracil replaces Thymine.



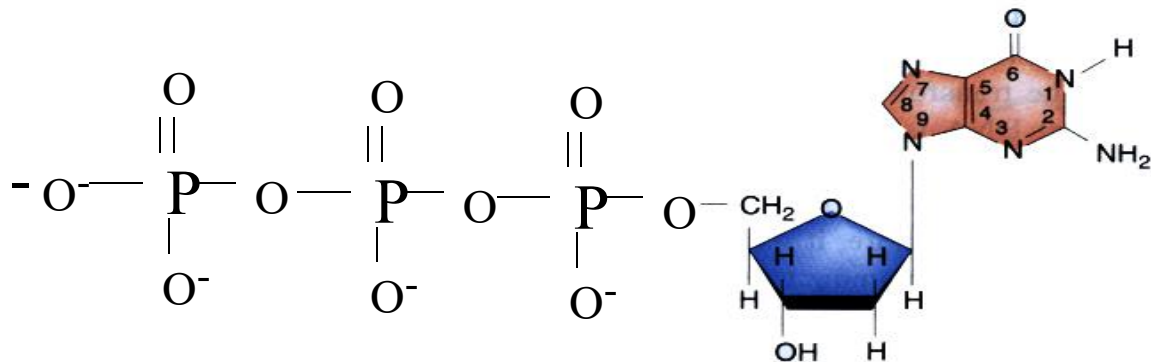
Ribonucleoside



dGMP

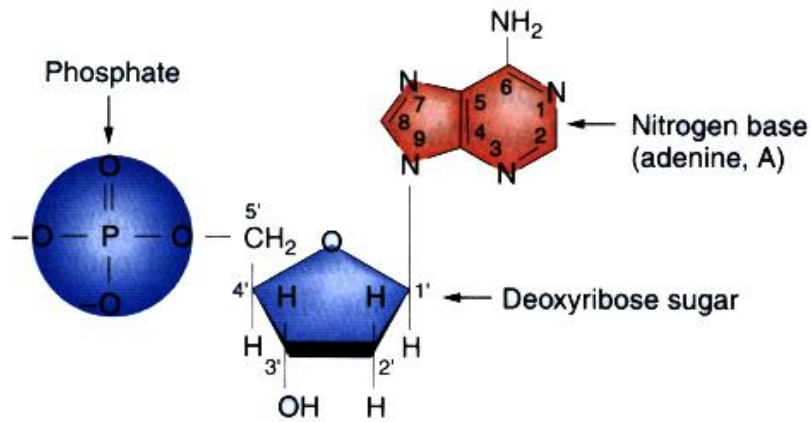


dGDP

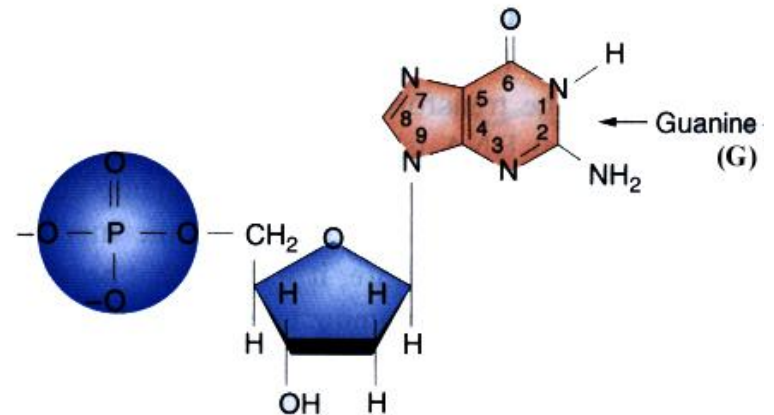


dGTP

Purine nucleotides

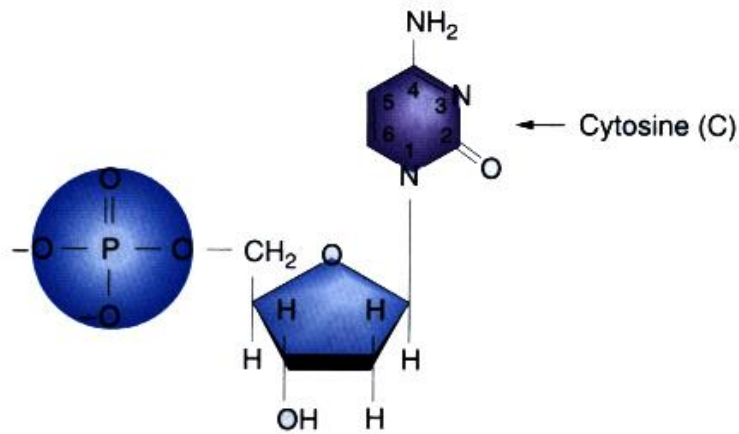


Deoxyadenosine 5'-phosphate (dAMP)

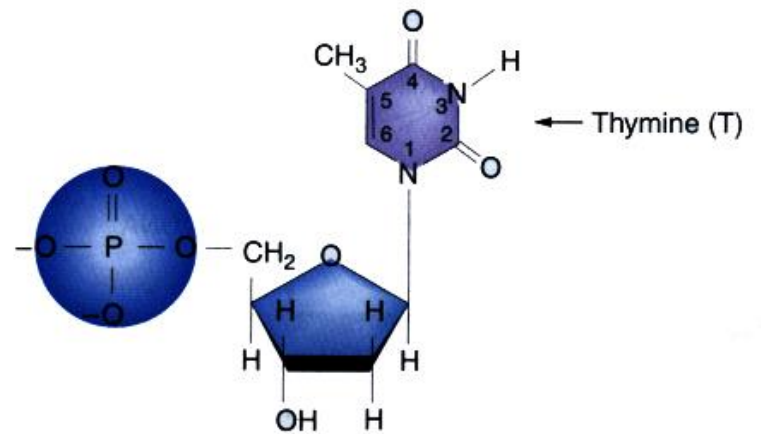


Deoxyguanosine 5'-phosphate (dGMP)

Pyrimidine nucleotides

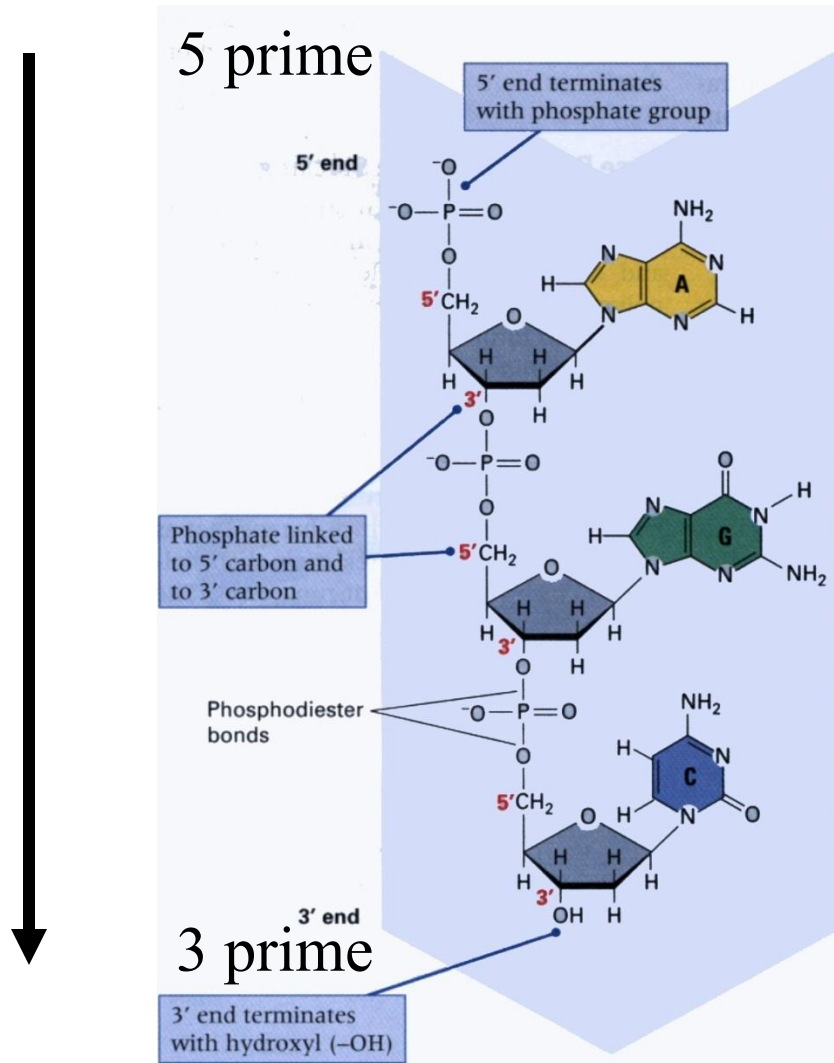


Deoxycytidine 5'-phosphate (dCMP)



Deoxythymidine 5'-phosphate (dTMP)

Single Strand of DNA

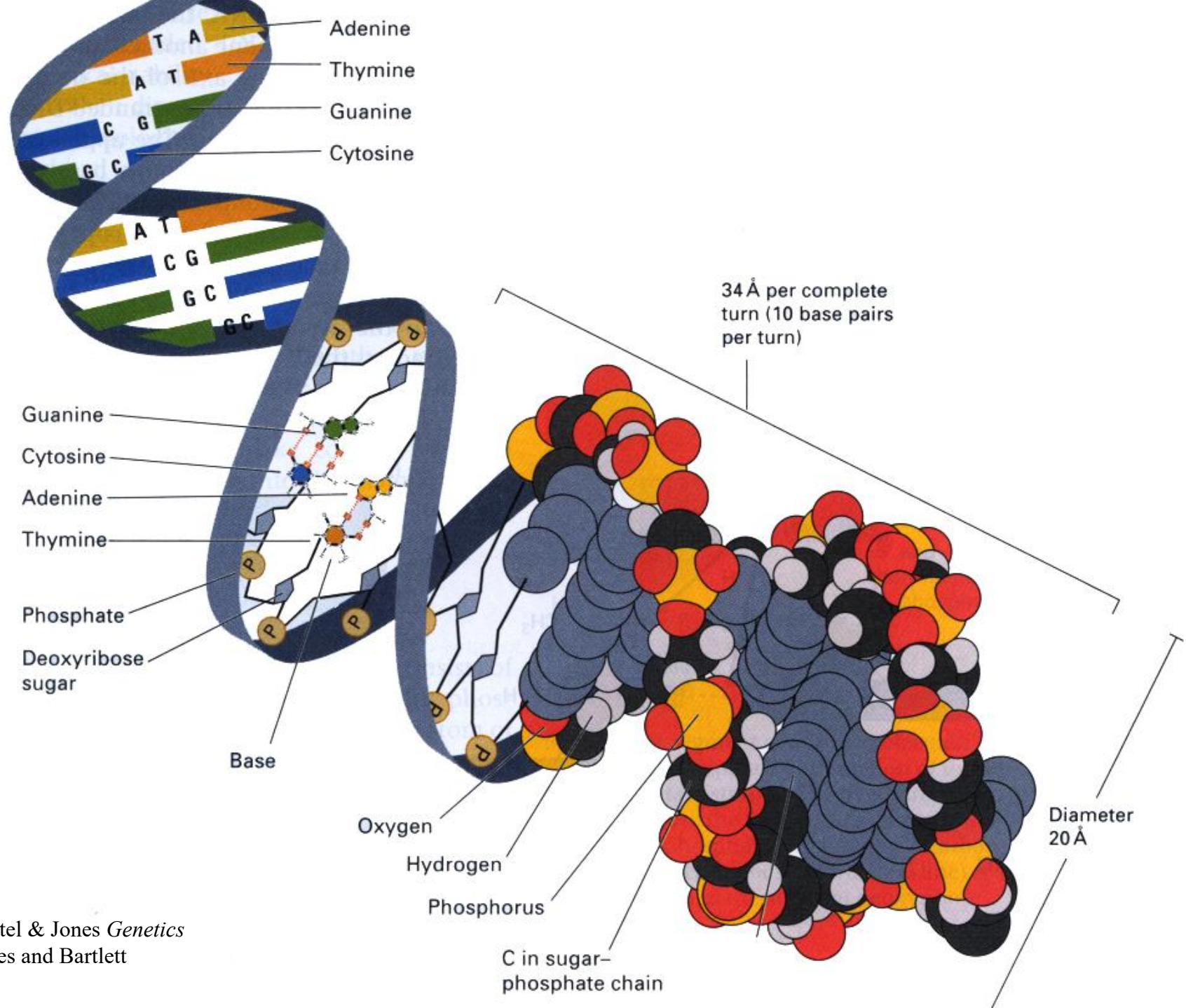


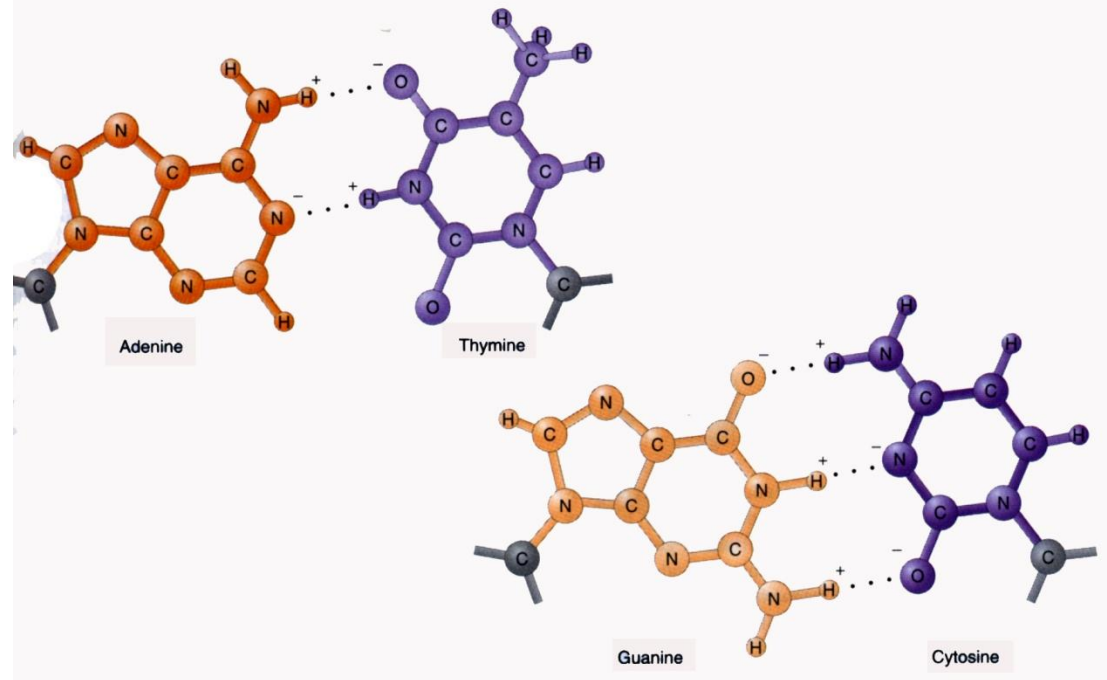
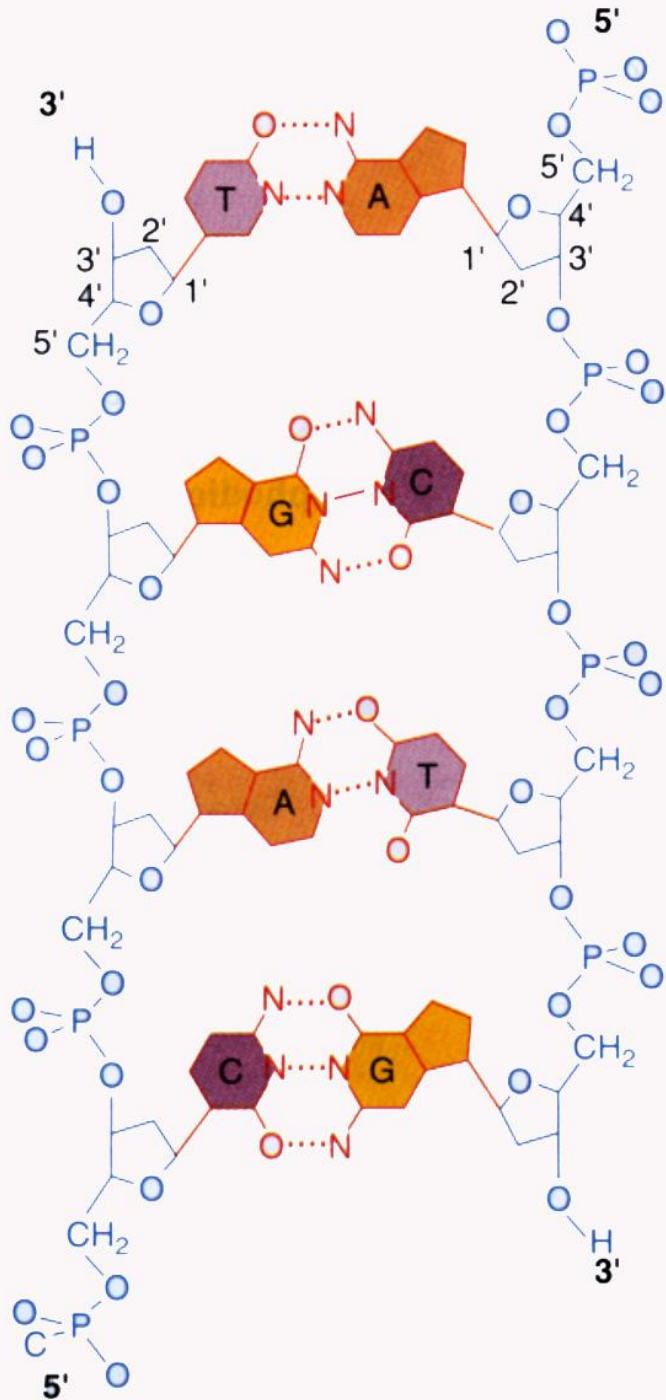
**-Has direction or polarity
-5' to3''**

**-Backbone comprised of
charged Phosphate residues**

**-Sugars Linked by a
phosphodiester bond**

**- phosphodiester bond Link the
5 ' methyl group to
the 3' OH group**

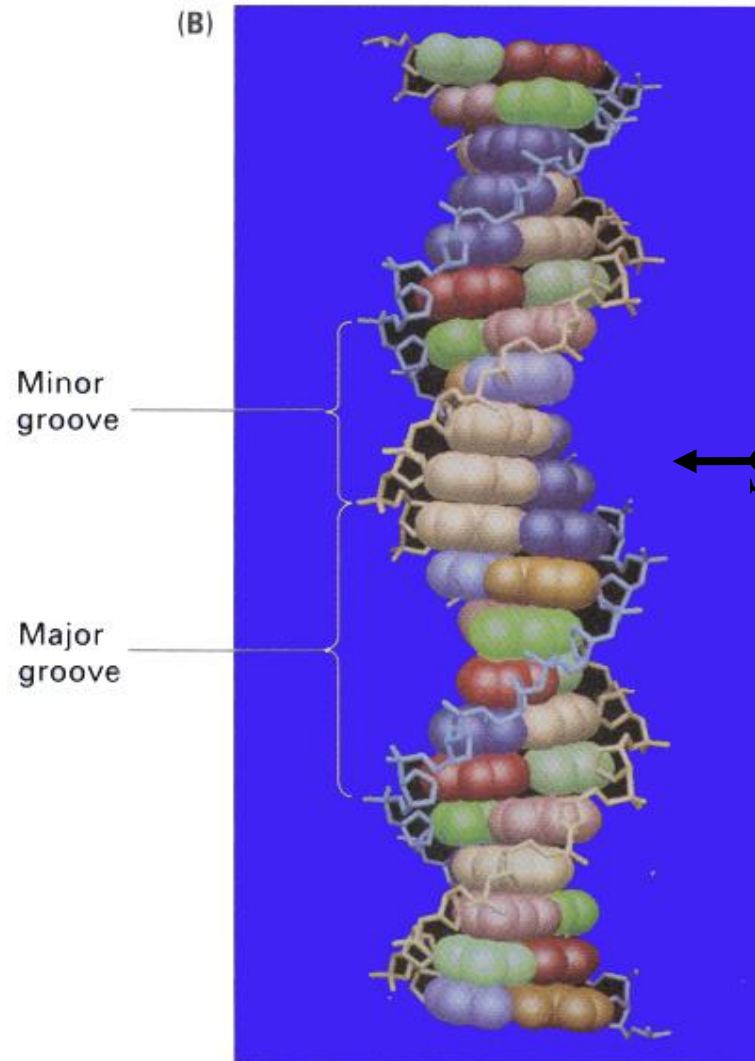
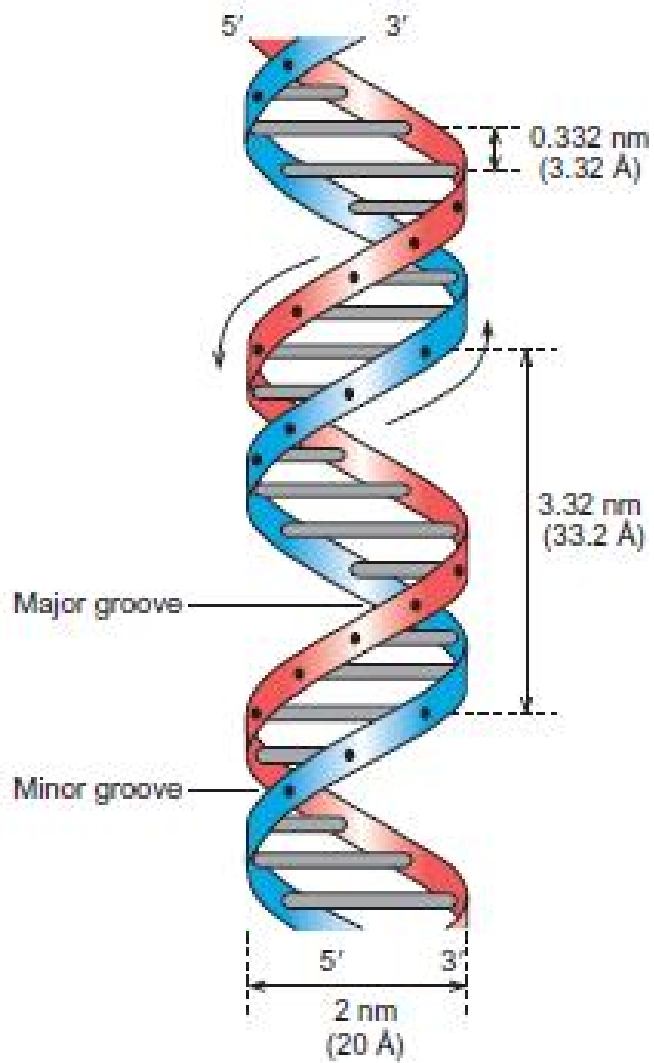




5'-3' phosphodiester bonds

5': free phosphate group

3': free hydroxyl group

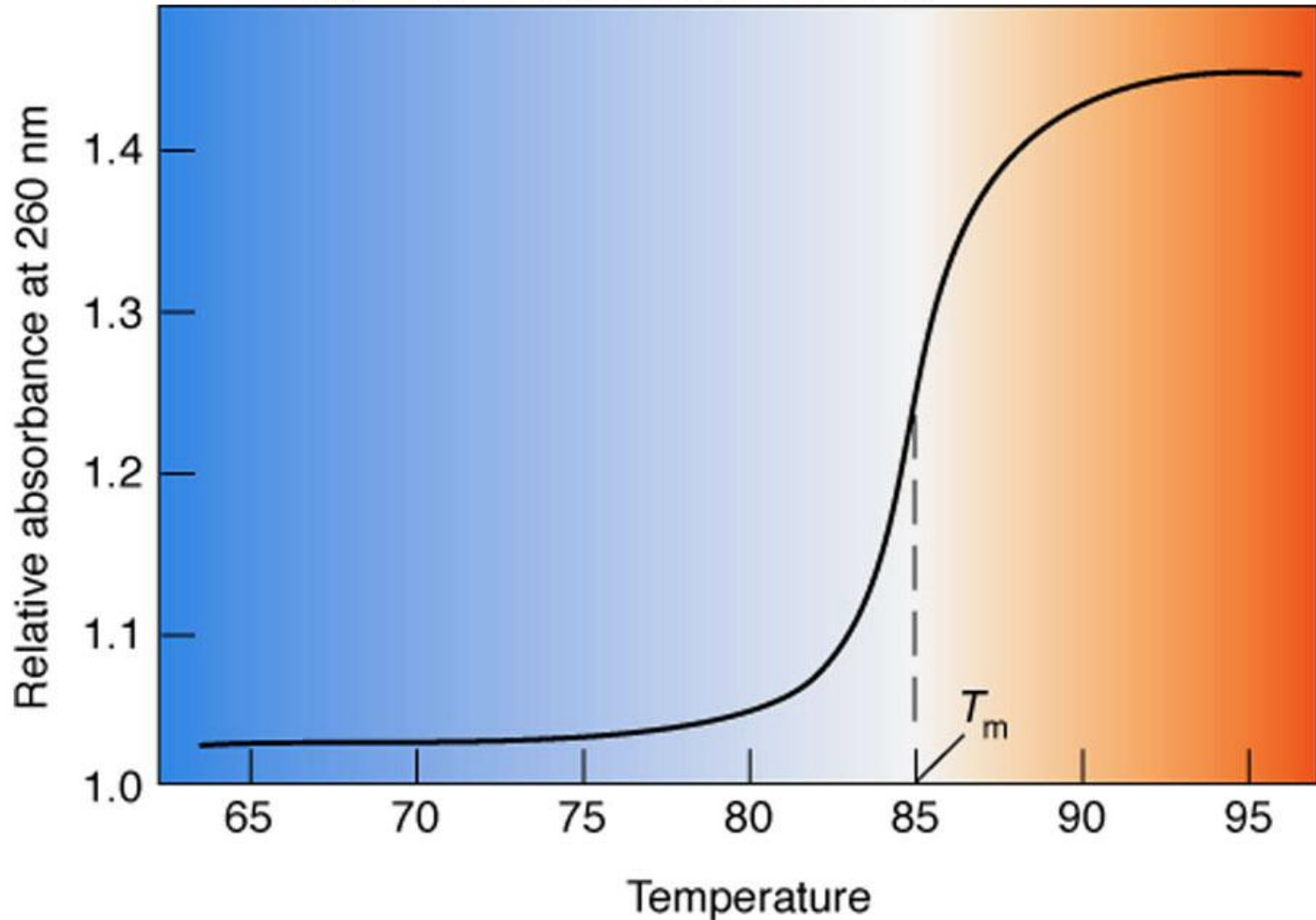


RNA differs from DNA

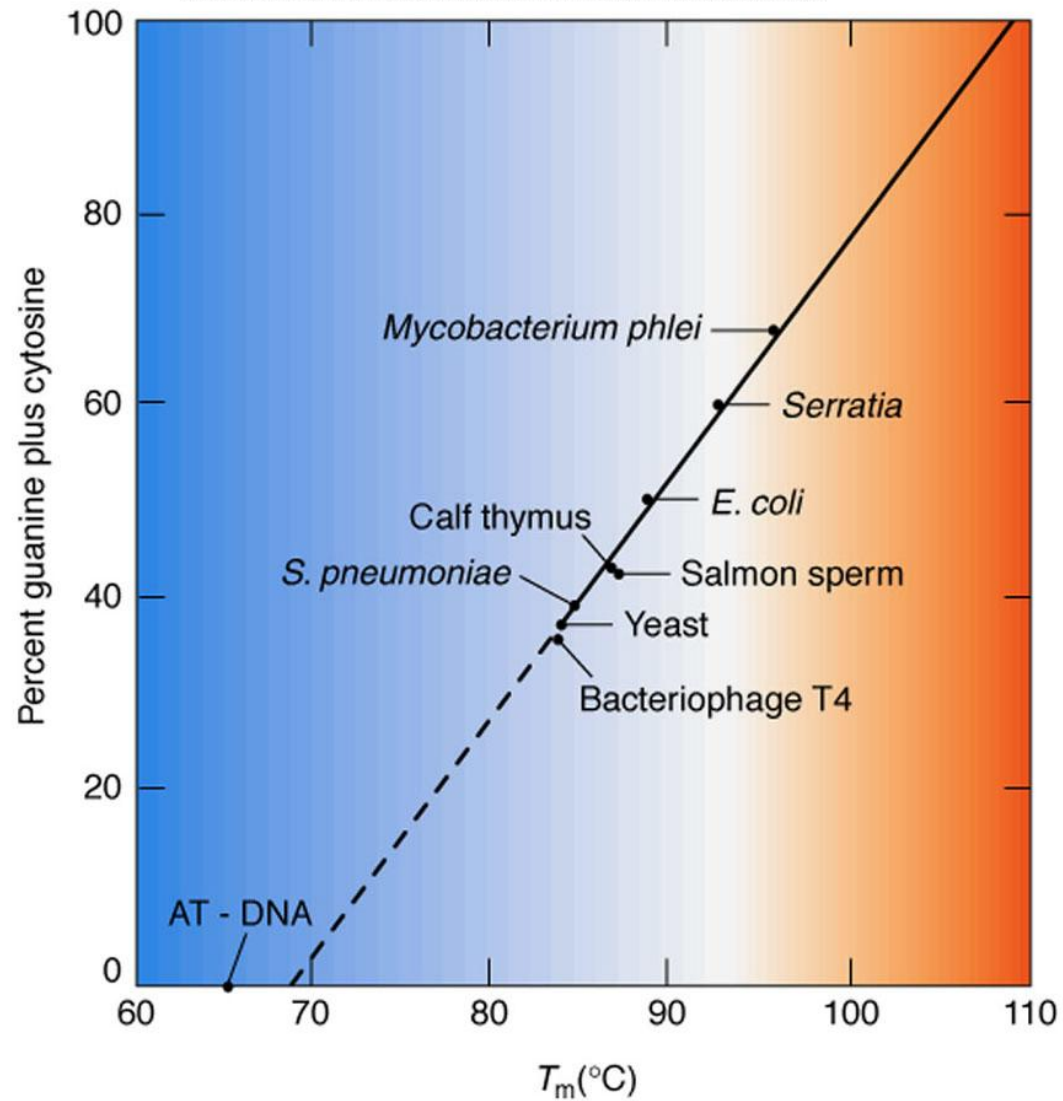
1. RNA has a sugar ribose
DNA has a sugar deoxyribose
2. RNA contains uracil (U)
DNA has thymine (T)
3. RNA molecule is single-stranded
DNA is double-stranded
4. RNA molecule is much smaller than DNA
5. Certain viruses contain genes made of RNA instead of DNA

Physical Chemistry of Nucleic Acids

- **DNA denaturation (melting)**: the doubled strands of DNA completely separated each other.
- Nucleic acids absorb UV light at 260nm because of the electronic structure in their bases, but when two strands of DNA come together, the close proximity of the bases in the two strands quenches some of this absorbance. When the two DNA strands separate, this quenching disappears and the absorbance rises 30-40%.
- **T_m** : the temperature at which the shift in absorbance is half completed.
- More GC, higher T_m



The DNA was heated, and its melting was measured by the increase in absorbance at 260nm. The point at which the melting is half complete is the melting temperature, or T_m . The T_m for this DNA under these condition is about 85

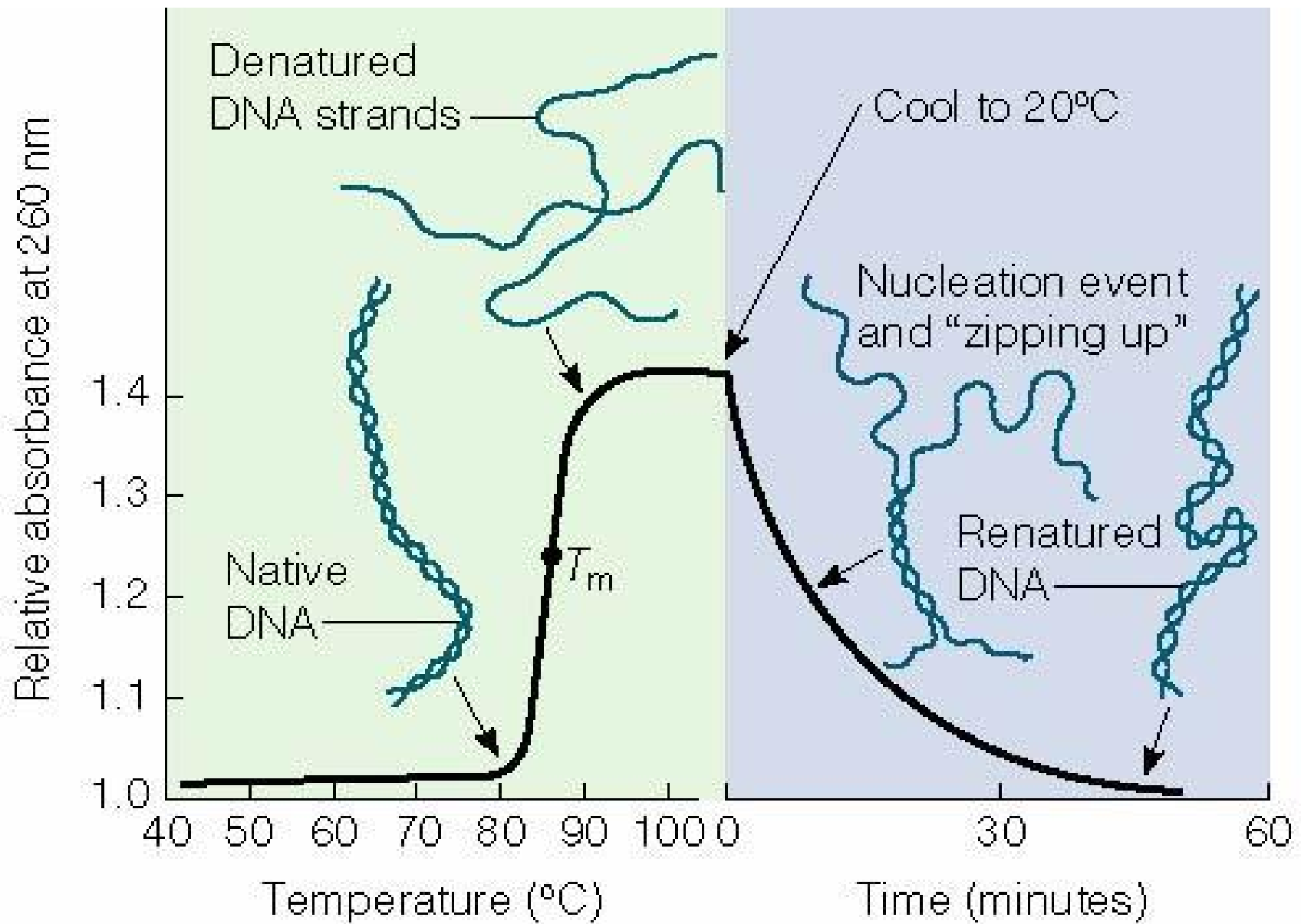


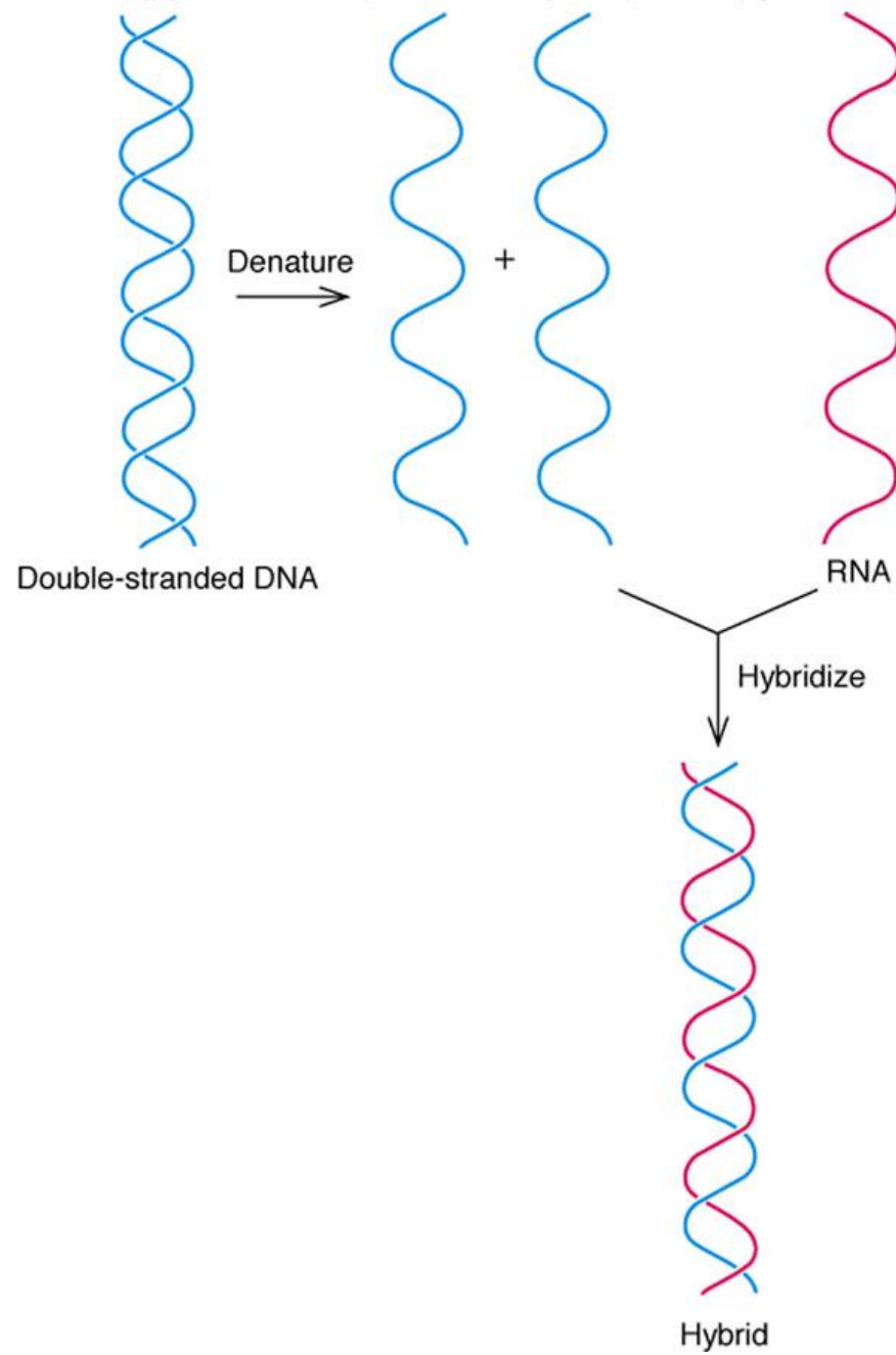
Relationship between DNA melting temperature and GC content.

DNA renaturation

- 1960, Julius Marmur et al. found:

when slowly cooled a denatured DNA solution, DNA regained the properties of the double helix.





Gene & Genome

- **Genes:**

- In non-molecular terms, a unit of inheritance that governs the character of a particular trait.
- In molecular terms, a segment of DNA containing the information for a single polypeptide or RNA molecule, including transcribed but non-coding regions.

- **Genome:** the complement of genetic information unique to each species of organism equivalent to the DNA of a haploid set of chromosomes from that species.

Non-nuclear genome

- The mitochondria (线粒体) and chloroplasts(叶绿体) also have a DNA genome (or chromosome).
- These resemble procaryotic genomes (likely due to the endosymbiotic origin of these organelles) but are much smaller.
-
- The mitochondrial genome varies in size among eukaryotes
 - ◆ mammals =16.5 kb & 37 genes
 - ◆ yeast and plants are greater than 5X this
- Chloroplasts are ~120 kb and have ~120 genes.

Genome outside the nuclear

Organelle DNA: maternally inherited; mainly circular; DNA is not complexes with histones

- **Mitochondria**: Site of cellular respiration, has its own genome and protein synthesis machinery
 - 线粒体是真核细胞中的细胞器。每个细胞中含有几十至数千个线粒体。每个线粒体有多个线粒体基因组拷贝。线粒体是非孟德尔式遗传方式，在多数高等生物中具有母性遗传的特征。
- **Chloroplast**: Site of photosynthesis, has its own genome and protein synthesis machinery
 - 叶绿体基因组是绿色植物特有的核外遗传结构。在多数高等植物中具有母性遗传的特征。
 - Non-Mendel inheritance.
 - Circular DNA molecular.
 - 120—217kb
 - Each chloroplast contain 1- tens of genomes

Genome size

DNA Content of Haploid Genomes of a Range of Phyla

C value paradox

- Size of haploid genome = **C-value**
- **C-value paradox**: maximum C-values suggest that some less evolved organisms may be more complex than more evolved organisms
- C-value paradox is explained ,not by extra genes, but by extra noncoding DNA in some organisms.

